JONAS EMILIO PETERSEN

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Computational engineer (Cambridge, Imperial) with 5 published papers (3 accepted at ICML 2026, incl. spotlight). Co-founded an AI startup scaled to >€500k revenue in 12 months. Now leading research at a $5M pre-seed deeptech startup building world models for industrial systems. Ex-BCG.

SKILLS & INTERESTS

AI/ML: Python, PyTorch, Transformers, time-series foundation models, LLMs, knowledge graphs
Miscellaneous: German (native), English (fluent), tennis, chess, jazz, hiking, robotics (SO-101, human2robot)

WORK EXPERIENCE

ETH Zürich — Zurich, Switzerland
Researcher, D-MAVT (part-time, under Prof. Siegwart) — *Feb 2026 – Present*
- Funded by Gebert Rüf Stiftung to accelerate Forgis' world model research for industrial systems

Forgis — Zurich, Switzerland
Head of Research — *Oct 2025 – Present*
- First employee at $5M pre-seed deeptech startup building HYPERION, a world model for industrial systems
- Defined research direction with CTO; actively partnering with 5 of the world's top-10 universities
- Assembled and leading a 10-person research team (5 MSc students, 2 PhDs, 1 postdoc, 2 professors)
- 3 first-author papers in 6 months; 3 accepted at ICML 2026 workshops (incl. spotlight)

Media Press Group — London, UK
CEO UK (prev. Process Automation Engineer) — *Sep 2023 – Sep 2025*
- Launched UK subsidiary, scaled to 6 people; Python + N8N sales automations, productised AI

K2 AI — London, UK
Co-founder — *Sep 2023 – Dec 2024*
- Grew to 10 employees, 3 hackathons, 3 AI-driven automation projects, >€500k revenue
- Conceptualised 4 AI-powered products; positive feedback from 30+ CFOs (100+ employee companies)
- Developed a task mining SaaS MVP using SLMs; reintegrated part of the team into parent organisation

Boston Consulting Group — Munich, Germany
Visiting Associate — *Jun – Aug 2021*
- Built a PowerBI costs forecasting tool; received full-time offer

Basler AG — Hamburg, Germany
R&D Intern — *Aug 2017*
- Python firmware unit tests & object detection with OpenCV

EDUCATION

University of Cambridge — Cambridge, UK
MPhil, Nanotechnology – Distinction (highest honours) — *Sep 2022 – Jul 2023*

Imperial College London — London, UK
MEng, Mechanical Engineering – Distinction, Dean's List (top 10%) — *Sep 2018 – Jul 2022*
- Robotics, Machine Learning, Advanced Computing, Mechatronics, Statistics, Embedded C
- Thesis on wildfire suppression optimisation via MCTS; published in *Combustion Science & Technology*

PUBLICATIONS

- **HEPA**: A Self-Supervised Horizon-Conditioned Event Predictive Architecture for Time Series — arXiv
  J. Petersen, G.-A. Lombardi, et al. | *Spotlight*, FMSD @ ICML 2026
- **FactoryNet**: A Large-Scale Dataset toward Industrial Time-Series Foundation Models — arXiv
  K. Othman, **J. Petersen**, et al. | AI4Physics @ ICML 2026 | FMSD @ ICML 2026
- **FactoryBench**: Evaluating Industrial Machine Understanding — arXiv
  Y. Merzouki, C. Izquierdo, ..., **J. Petersen**, P. Petersen | Under review
- **Tabula RASA**: What Structural Inductive Bias Helps Transformers Reason Over Knowledge Graphs? — arXiv
  J. Petersen, C. Mazzoleni, et al. | GFM @ ICML 2026
- **MCTS**: Modelling and Optimisation of Extinction Actions for Wildfire Suppression — PDF
  J. Petersen, S. Kapur, S. Gkantonas, E. Mastorakos, A. Giusti | *Combustion Science & Technology* (2023)